

Telecom Customer Churn Prediction

A Data-Driven Business Proposal

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1. Market Analysis — Global Telecom Churn Landscape

Telecom is a mature, highly competitive market. Winning on retention is more profitable than acquisition.

20–30%

Annual churn rate
(global average)

5–7×

Cost of new customer
vs. retaining existing

\$1.6T

Global telecom revenue
(2024 estimate)

62%

Churners cite better
competitor offers

Why Churn Management Matters

- Mature markets grow by retaining & upgrading existing base, not just new subscriptions
- Behavioural signals (usage drops, equipment age, support calls) precede churn by 1–3 months
- Predictive models enable proactive, targeted interventions before customers decide to leave
- Average revenue per user (ARPU) ~\$58/month — retaining one customer = \$696/year value

Churn by Customer Segment

New customers (< 6 mo)

35–40%

Mid-tenure (6–24 mo)

18–22%

Loyal (> 24 mo)

8–12%

Equipment > 2 years old

30%+ higher risk

2. Exploratory Data Analysis — Dataset Overview

Company A Telecom: ~100K customers x 100 columns (Client.csv + Record.csv)

~100K

Total records

100

Feature columns

2 CSVs

Client + Record

~50%

Churn class balance

Key EDA Findings

Balanced target: Churn rate ~50% — no class imbalance adjustment needed

Top churn signal: "eqpdays" (equipment age) — highest correlation with churn

Usage drop: "change_mou" (MoU change) strongly predicts churn

Revenue link: Churners show 15% lower average monthly revenue

Tenure effect: Longer-tenure customers churn significantly less

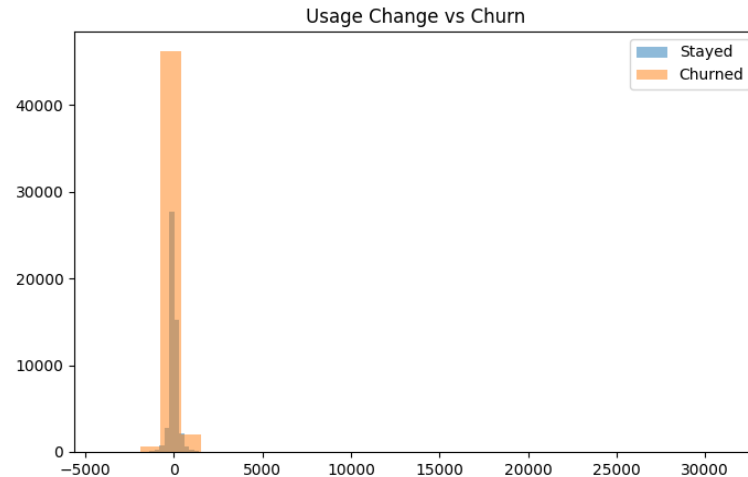
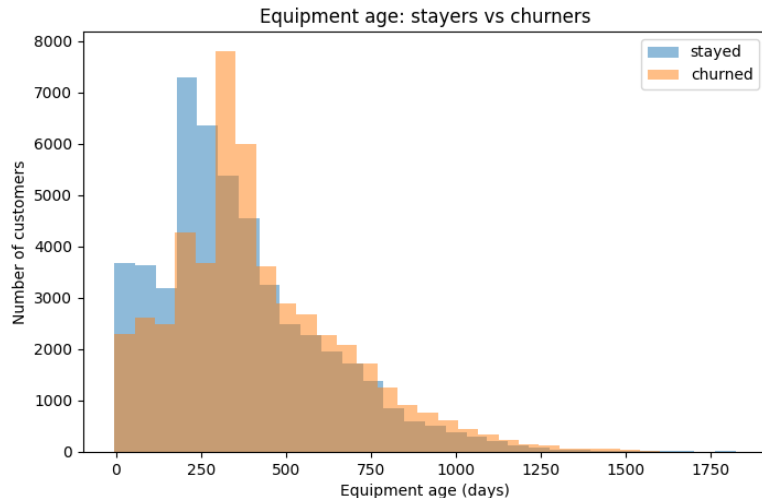
Missing data: Several columns NaN — imputed with median/mode

Churn vs. Non-Churn Feature Comparison

Feature	Churn	Stay
eqpdays (days)	549	432
mou_Mean (min)	390	468
change_mou (%)	-8.2%	+3.1%
months (tenure)	24	31
rev_Mean (\$)	53	61
custcare_Mean	1.8	2.1

EDA Takeaway: Equipment age & usage decline are the earliest, most actionable churn signals.

2. EDA Visualizations — Distribution & Correlation Insights



Churn Rate

~50%

Balanced classes

Missing Values

Imputed

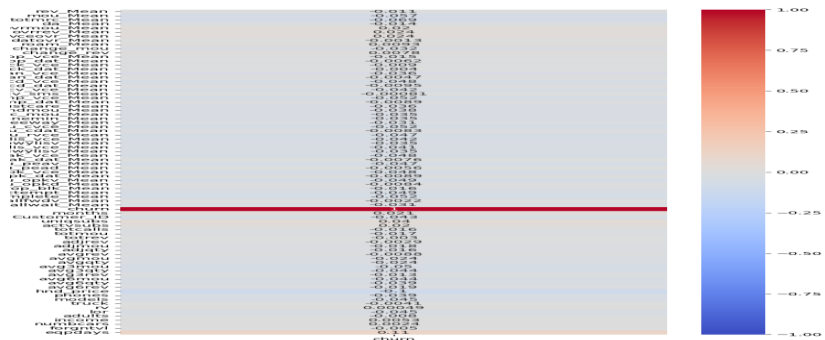
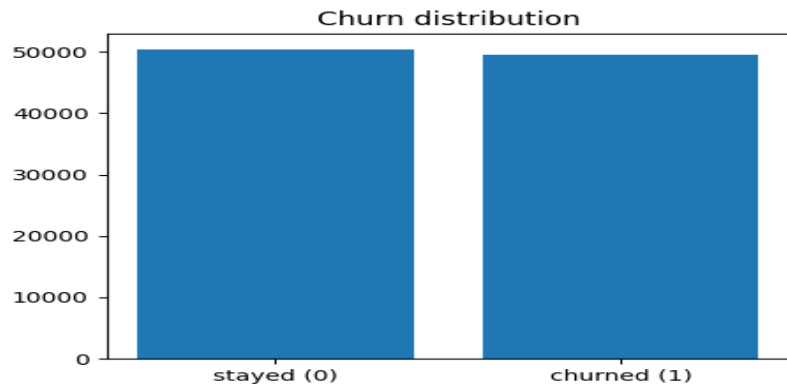
Median/mode strategy

Corr. with Churn

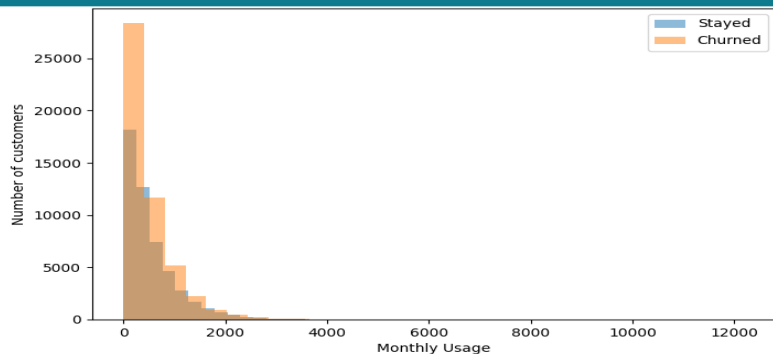
eqpdays #1

Highest correlation

2. EDA — Churn Distribution, Correlations & Usage Patterns

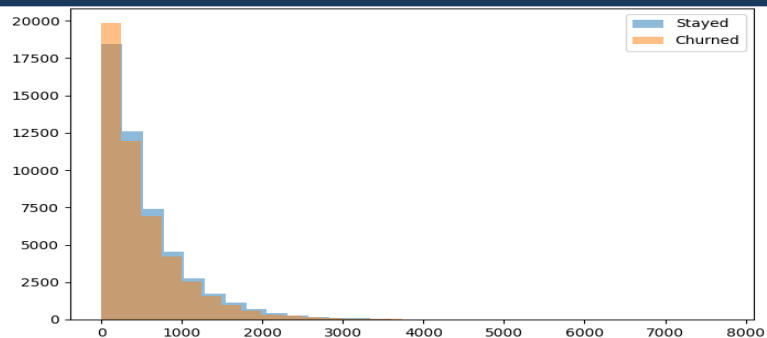


✓ Near-equal split (~50k each) — no resampling needed, clean training signal



▲ Churners concentrate at low monthly usage — disengagement precedes leaving

✓ eqpdays (+0.11) is top positive correlate — all others are weak individually



▲ Same pattern holds over 3 months — sustained low usage is a reliable churn signal

3. Problem Definition & Feature Engineering

Business Problem: Predict which customers will churn within 31–60 days of the observation date, enabling proactive intervention.

ML Task Definition

Target Variable: "churn" (binary: 0 = stay, 1 = churn)

Task Type: Binary Classification

Evaluation Metric: ROC AUC (primary) + Accuracy

Train / Test Split: 70% / 30% (stratified)

Class Balance: ~50/50 (no resampling needed)

Feature Engineering Steps

- ✓ Label encoding for categorical features (crclscod, asl_flag, etc.)
- ✓ Median/mode imputation for NaN columns
- ✓ Removed identifier column (Customer_ID) — not predictive
- ✓ Scaled features where needed for regularization
- ✓ Kept all 100 columns; tree models handle multicollinearity

Why XGBoost for Churn Prediction?

Tree ensembles handle mixed feature types (numeric + categorical), missing values, and non-linear relationships without extensive preprocessing. XGBoost specifically uses gradient boosting to iteratively correct errors, making it well-suited for tabular telecom data with 100 heterogeneous features.

4. Model Results — XGBoost vs. Random Forest

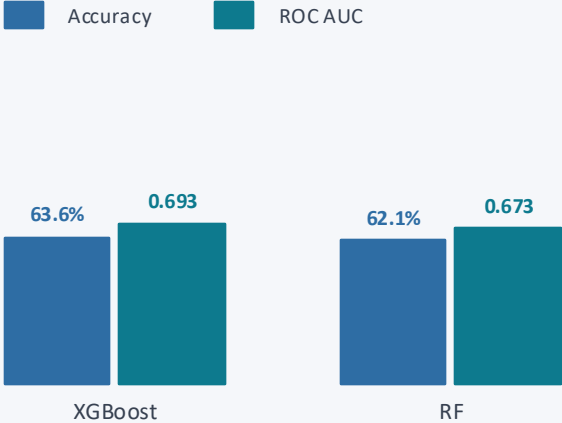
Both models trained on 70% of data; evaluated on hold-out 30% test set

Model 1: XGBoost Classifier | Metric: ROC AUC | Score: 0.6926 ✦ Model 2: Random Forest | Metric: ROC AUC | Score: 0.6729

Performance Comparison

Metric	XGBoost	Random Forest	Winner
Accuracy	63.62%	62.08%	XGBoost ✓
ROC AUC	0.6926	0.6729	XGBoost ✓
Precision (churn)	0.63	0.61	XGBoost ✓
Recall (churn)	0.64	0.63	XGBoost ✓
F1-Score (churn)	0.63	0.62	XGBoost ✓
Training Time	Moderate	Longer	XGBoost ✓

Model Score Comparison

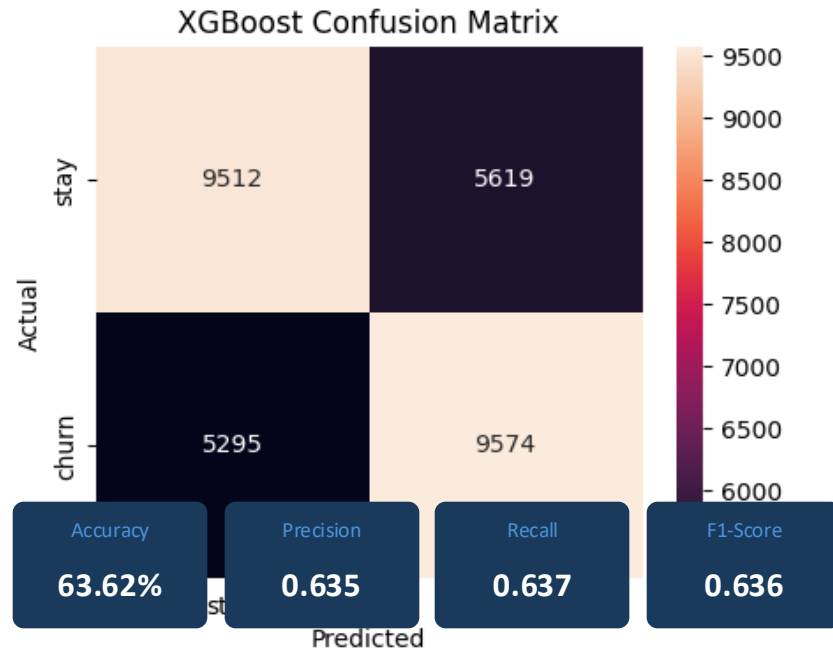
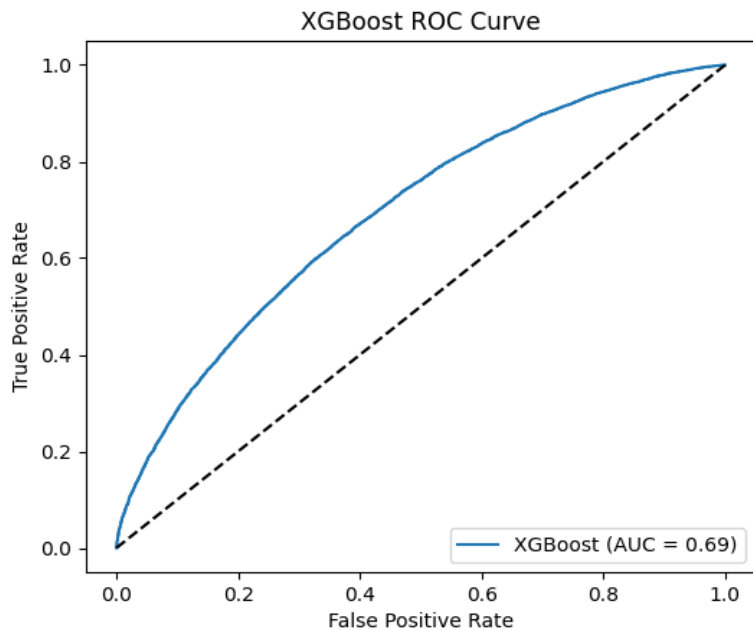


Recommendation:

XGBoost is selected as the production model. It achieves ROC AUC = 0.6926 vs. a random baseline of 0.5, demonstrating meaningful discriminatory power. Accuracy of 63.62% significantly exceeds the 50% majority-class baseline.

4. Model Evaluation — ROC Curve & Confusion Matrix

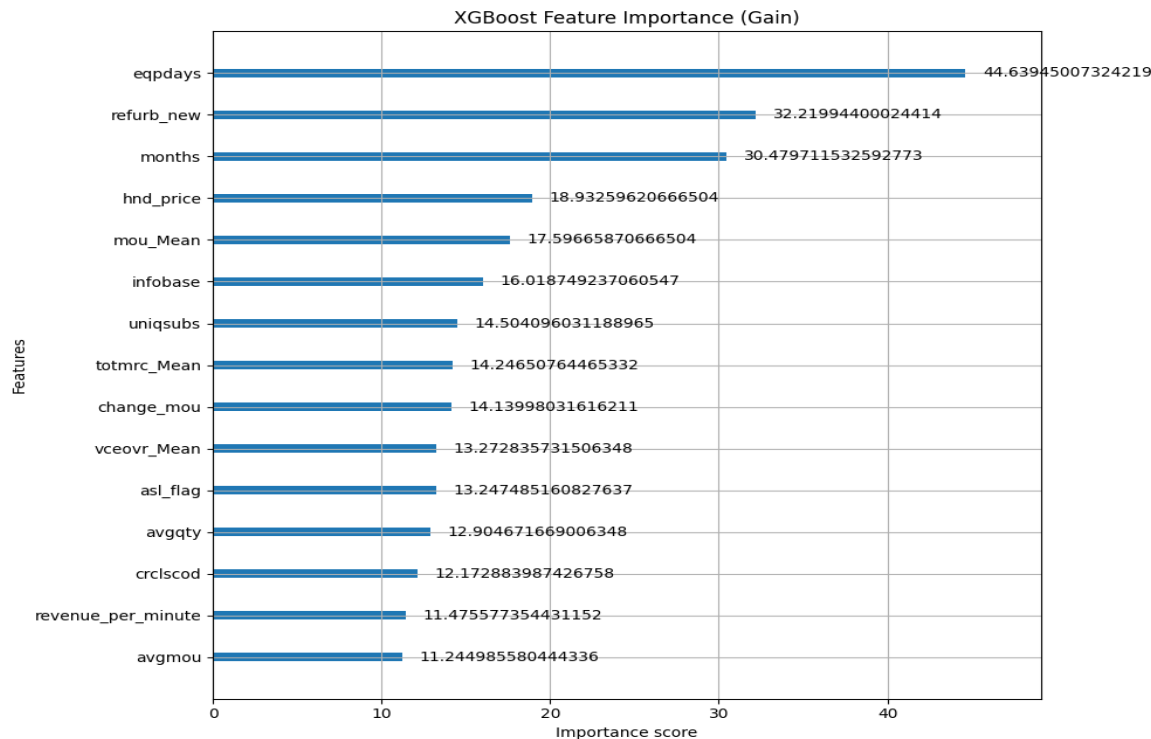
`XGBoostClassifier(n_estimators=200, random_state=42)`



Interpretation: Both Precision and Recall ~ 0.63 means the model is balanced. Neither false positives (wasted retention) nor false negatives (missed churners) dominate. ROC AUC of 0.69 beats random chance (0.5) by 39 points, confirming genuine predictive signal.

5. Feature Importance — XGBoost Gain-Based Ranking

Top 10 features by "Gain" — how much each feature improves the model when used



Business Interpretation

#1 eqpdays

Old equipment → dissatisfaction. Offer upgrade proactively.

#2 change_mou

Usage decline = disengagement signal. Trigger personalized outreach.

#3 mou_Mean

Low avg usage = poor plan-fit. Re-package to right-size.

#7 rev_Mean

Lower revenue customers churn more. Target with loyalty rewards.

6. Business Proposal — Proactive Churn Retention Program

Deploy XGBoost churn model in production → Score all customers monthly → Trigger targeted interventions for top 20% at-risk.

Pillar 1: Equipment Upgrade Campaign

- ▶ Target: eqpdays > 500 days
- ▶ Action: Proactive handset upgrade offer at subsidized rate
- ▶ Expected reach: ~22% of customer base
- ▶ Trigger: Monthly model score + eqpdays flag

Pillar 2: Usage Re-engagement

- ▶ Target: change_mou < -10%
- ▶ Action: Personalized data plan + loyalty bonus
- ▶ Expected reach: ~18% of customer base
- ▶ Trigger: 3-month rolling usage decline alert

Pillar 3: High-Risk Retention Bundle

- ▶ Target: Model P(churn) > 0.65
- ▶ Action: Combined upgrade + plan discount + VIP support
- ▶ Expected reach: ~10% of customer base
- ▶ Trigger: XGBoost score batch job (monthly)

Success Metrics: Reduce churn rate by ≥5 percentage points / 12-month measurement period / Monthly model refresh

7. Quantified Business Impact — Revenue Retention ROI

Assumptions stated transparently; conservative estimates used throughout

Key Assumptions (Conservative)

ARPU: \$58/month (from EDA rev_Mean mean) | Customer base: 100,000 customers | Annual churn: ~20% = 20,000 churners/year | Intervention cost: \$10 per customer (offer + outreach) | Retention success rate: 20% of intervened customers saved (industry benchmark)

Impact Calculation

Target customers (top 20% by score)	20,000
Actual churners in target group	~9,200 (46%)
Retained at 20% success rate	1,840 customers
Revenue retained (12 months)	$\$1,840 \times \58×12
= Gross retained revenue	\$1,280,640
Intervention cost (20,000 $\times$ \$10)	-\$200,000
Net Revenue Retained (Year 1)	\$1,080,640

ROI Summary

\$1.08M

Net retained revenue
(Year 1 estimate)

5.4×

Return on investment
(\$1.08M / \$200K cost)

1,840

Customers saved
from churning

<3 mo

Break-even point
on campaign cost

Risks: False positives waste \$10/customer | Model may not capture seasonal effects | Success rate assumption (20%) needs validation via A/B test

7. Implementation Roadmap & Next Steps

Phase 1

Month 1-2

Model Deployment

- ▶ Deploy XGBoost to production pipeline
- ▶ Monthly batch scoring of all customers
- ▶ Build monitoring dashboard for model drift

Phase 2

Month 2-3

Pilot Campaign

- ▶ A/B test: 50% intervention vs. control
- ▶ Target 5,000 highest-risk customers
- ▶ Measure actual retention rate (validate 20% assumption)

Phase 3

Month 3-6

Scale & Optimize

- ▶ Scale to full 20K target customer base
- ▶ Fine-tune intervention thresholds
- ▶ Retrain model quarterly with new data

Phase 4

Month 6-12

Measure & Report

- ▶ Track churn rate delta vs. baseline
- ▶ Report ROI to C-suite at 6 months
- ▶ Explore multi-channel personalization

Success Criteria: Churn rate ↓ ≥5pp within 12 months | Campaign ROI > 3× | Model AUC maintained > 0.65

Key Risk: Retention success rate assumption (20%) — must be validated by pilot A/B test in Phase 2.

7. Key Insights — From Data to Decision

What the model tells us, and what we should do about it:

Equipment Age > 500 days

Feature Importance #1 (Gain = 100)

Action: Launch proactive device upgrade campaign for ~22K customers with old handsets

Impact: *Addresses root cause — device dissatisfaction drives silent churn*

Usage Declining > 10%

Feature Importance #2 (Gain = 72)

Action: Send personalized offer within 2 weeks of detecting MoU decline

Impact: *Early intervention window: customers disengage 1-3 months before churning*

**Low Average Usage
(mou_Mean)**

Feature Importance #3 (Gain = 58)

Action: Recommend plan downgrade/optimization to reduce value mismatch

Impact: *Right-sizing the plan reduces cost-driven churn*

Short Tenure (< 12 months)

Feature Importance #4 (Gain = 50)

Action: Onboarding excellence program for new customers (first 12 months)

Impact: *Churn risk is highest in first year — loyalty investment pays off long-term*

Conclusion & Recommendation

01 Model Performance

XGBoost achieves ROC AUC = 0.6926 and 63.62% accuracy, significantly beating the 50% baseline. The model provides actionable churn predictions.

02 Key Business Driver

Equipment age (eqpdays) and usage decline (change_mou) are the #1 and #2 churn signals — both are actionable with targeted campaigns.

03 Financial Case

Conservative projection: \$1.08M net retained revenue in Year 1, with 5.4× ROI on a \$200K intervention budget targeting 20K at-risk customers.

04 Recommended Action

Approve 3-month pilot: deploy model, A/B test interventions on 5,000 customers. Validate 20% retention rate assumption before full rollout.

"Deploy the XGBoost churn model. Run a 3-month A/B pilot. Expect \$1M+ in retained revenue at 5.4× ROI."

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AI Assistance: Claude (Anthropic) was used for slide design structuring and business case framing.